

Samples

This page describes the sample content shipped under:

- `Assets/MIDI/Samples/`

Scenes

Location: Assets/MIDI/Samples/Scenes/

- `MidiSampleScene.unity`
Demonstrates MIDI 1.0 receive/send workflows via `MidiManager` and MIDI 1.0 event handler interfaces.
Includes an optional toggle to route output into an MPTK-backed virtual output device (when enabled).
- `Midi2SampleScene.unity`
Demonstrates MIDI 2.0 / UMP workflows via `Midi2Manager` and MIDI 2.0 handler interfaces.
Includes an optional toggle to mirror MIDI 2.0 sends into an MPTK MIDI 1.0 virtual sink (when enabled).
- `SampleMenuScene.unity`
Hub scene that lists and opens other sample scenes (Window > MIDI > Samples/Open Sample Menu Scene).

Unity Ecosystem Integration

Location: Assets/MIDI/Samples/Integrations/Scenes/

- `MidiAnimatorIntegrationSampleScene.unity`
Demonstrates MIDI → Animator parameter driving via `MidiAnimatorDriver` / `MidiBlendTreeDriver` .
CC1 changes the cube's height, CC10/11 rotate it, and Note 60 changes the pulse effect.
- `MidiTimelineIntegrationSampleScene.unity`
Demonstrates Timeline-synchronized SMF playback via `PlayableDirector` + `SmfPlayer` + `MidiPlaybackTrack` .
During Play Mode, you can control Timeline Play / Pause / Seek from IMGUI.
- `MidiVisualScriptingIntegrationSampleScene.unity`
Demonstrates Visual Scripting integration via `MidiVisualScriptingBridge` and Event Bus listeners.
You can add **Events > MIDI** nodes to the Script Graph to extend it on the same GameObject.

Gameplay

Location: Assets/MIDI/Samples/Gameplay/

- `Scenes/MidiGameplaySampleScene.unity`
Demonstrates the wiring and behavior of the gameplay components (`MidiDeviceFilter` → `MidiChannelFilter` → `MidiInputRouter` / `MidiNoteTracker`).
During Play Mode, you can simulate Note / CC / MIDI Start / SysEx through a virtual device from the IMGUI panel. Physical MIDI controllers work the same way.
- `Scenes/MidiClockSyncSampleScene.unity`
External MIDI Clock injection, `MidiClockSync` BPM estimation, and `SmfPlayerClockAdapter` demo.
- `Scenes/ChordScaleSampleScene.unity`
`ChordRecognition` / `MidiChordDetector` demo.
- `Scenes/ChordPuzzleSampleScene.unity`
Target-chord quiz (`MidiChordDetector.targetChord`) demo.
- `Scenes/ScalePracticeSampleScene.unity`
Practice demo with multiple scale types.

Networking / Input System / Chunity

Location: `Assets/MIDI/Samples/Integrations/`

- `Networking/Scenes/MidiNetworkJamSampleScene.unity`
UDP-based `MidiNetworkHub` / `MidiNetworkClient` demo (`FEATURE_MIDI_NETWORK` required).
- `Networking/Scenes/MidiMirrorNetworkSampleScene.unity`
Mirror Host/Client MIDI (`FEATURE_MIDI_NETWORK` + `FEATURE_MIRROR` ; Mirror installed).
- `Networking/Scenes/MidiNetcodeNetworkSampleScene.unity`
Netcode Host/Client MIDI (`FEATURE_MIDI_NETWORK` + `FEATURE_NETCODE` ; NGO installed).
- `Networking/Scenes/MidiWsnet2NetworkSampleScene.unity`
WSNet2 Create/Join MIDI (`FEATURE_MIDI_NETWORK` + `FEATURE_WSNET2` ; WSNet2 client + running servers).
- `InputSystem/Scenes/InputSystemBridgeSampleScene.unity`
`MidiInputSystemBridge` + Synthetic Device demo (`FEATURE_INPUT_SYSTEM` required).
- `Chunity/Scenes/ChunityBridgeSampleScene.unity`
`MidiChuckBridge` + `MidiChuckPatchHost` demo (external Chunity + `FEATURE_CHUNITY` required).
- `Chunity/Scenes/ChunityWorkflowsSampleScene.unity`
Poly / SMF / Clock / Event→MIDI workflow demos (`FEATURE_CHUNITY` required).

- `Chunity/Scenes/ChunityPresetsSampleScene.unity`
MidiChuckPreset / MidiChuckParameterBinder with Filter / Gain, etc. demo
(`FEATURE_CHUNITY` required; Timeline optional via `FEATURE_USE_TIMELINE`).
- `Chunity/Scenes/ChunityMicFxSampleScene.unity`
Mic adc + PitShift FX, demo controlling the pitch-shift amount with CC1 (Mod Wheel)
(`FEATURE_CHUNITY` required; Phase E).
- `Chunity/Scenes/ChunityGeneratorWorkflowSampleScene.unity`
Main + Sub Generator + Bridge demo (`FEATURE_CHUNITY_SCRIPTABLE_AUDIO` required).

Scriptable Audio

Location: `Assets/MIDI/Samples/Integrations/ScriptableAudio/`

- `Scenes/ScriptableAudioMetronomeSampleScene.unity`
Metronome / DSP bootstrap demo (`FEATURE_SCRIPTABLE_AUDIO` , Unity 6000.3+).
- `Scenes/ScriptableAudioSequenceSampleScene.unity`
SMF sequence → DSP schedule demo.
- `Scenes/ScriptableAudioUmpSequenceSampleScene.unity`
UMP sequence → DSP schedule demo.

Foundation

Location: `Assets/MIDI/Samples/Foundation/`

- `Scenes/FoundationSampleScene.unity`
Device selection, latency calibration, and Foundation UI shell demo.

Maestro / MPTK

Location: `Assets/MIDI/Samples/MPTK/`

- `Scenes/MptkIntegrationSampleScene.unity`
Maestro / MPTK integration hub (`FEATURE_USE_MPTK` required; Pro features need `MPTK_PRO`).

MIDI Tracker (separate sample)

Location: `Assets/MIDITracker/` (outside `Assets/MIDI/Samples/` , depends on the core plugin one-way)

- `Scenes/MidiTrackerSampleScene.unity`
Pattern-oriented MIDI tracker showcase (edit, arrange, record, SMF I/O, multi-platform UX).
See [MIDI Tracker README](#) and [MIDI Tracker development plan](#).

Scripts

Location: `Assets/MIDI/Samples/Scripts/`

- `MidiSampleScene.cs`

Typical responsibilities in a demo:

- initialize MIDI
- register handler objects
- react to incoming messages (e.g., log note on/off)
- send test messages
- (optional) route output into MPTK via a virtual device

- `Midi2SampleScene.cs`

Typical responsibilities in a demo:

- initialize MIDI 2.0
- register UMP handlers
- display decoded events
- send example UMP messages
- (optional) mirror sends into MPTK via a MIDI 1.0 virtual sink

Unity Ecosystem Integration

Location: `Assets/MIDI/Samples/Integrations/Scripts/`

- `MidiAnimatorIntegrationSampleScene.cs`

Runtime construction of `MidiAnimatorDriver` / `MidiBlendTreeDriver`, virtual MIDI input, IMGUI control panel

- `MidiTimelineIntegrationSampleScene.cs`

Runtime construction of `TimelineAsset` / `MidiPlaybackTrack` / `SmfPlayer`, Timeline control UI

- `MidiVisualScriptingIntegrationSampleScene.cs`

Event Bus demo via `MidiVisualScriptingBridge` + `MidiVisualScriptingSampleFeedback`

- `MidiIntegrationSampleFactory.cs`

Shared helper for demo SMF arpeggio generation and virtual device registration

Gameplay

Location: `Assets/MIDI/Samples/Gameplay/Scripts/`

- `MidiGameplaySampleScene.cs`

Typical responsibilities in a demo:

- runtime construction of the filter chain (DeviceFilter → ChannelFilter)
- NoteOn / CC / MIDI Start / SysEx bindings via `MidiInputMap` / `MidiInputRouter`
- held-note tracking and chord detection via `MidiNoteTracker`
- hardware-free testing by injecting into a virtual input device (`virtual:gameplay-sample`)
- filter configuration and event-log display via ImGui
- `MidiClockSyncSampleScene.cs` — Clock sync demo
- `ChordScaleSampleScene.cs` — chord/scale detection demo
- `GameplaySampleFactory.cs` — virtual MIDI / Clock injection helper

Networking / Input System

Location: `Assets/MIDI/Samples/Integrations/`

- `Networking/Scripts/MidiNetworkJamSampleScene.cs` — LAN MIDI sync demo (`#if FEATURE_MIDI_NETWORK`)
- `InputSystem/Scripts/InputSystemBridgeSampleScene.cs` — Input System bridge (`#if FEATURE_INPUT_SYSTEM`)

Foundation

Location: `Assets/MIDI/Samples/Foundation/Scripts/`

- `FoundationSampleScene.cs` — foundation demo
- `FileUtility.cs` , `AudioClipUtility.cs` (under `Assets/MIDI/Samples/Scripts/`)
Utility scripts used by the samples (file handling, audio clip helpers).

WebGL templates

Location: `Assets/MIDI/Samples/WebGLTemplates/`

Contains WebGL build templates intended for MIDI-enabled WebGL deployments. Use these if you want consistent HTML/JS scaffolding for WebGL MIDI and BLE MIDI flows.

Recommended workflow to test quickly

1. Open one of the sample scenes.
2. Enter Play Mode.
3. Connect a MIDI device (or use a network transport like RTP-MIDI / UDP MIDI 2.0).
4. Confirm:
 - device attach events appear,
 - note/CC events are received,
 - sending produces output on the target device.

MIDI 1.0 Basics (MidiSampleScene)

1. Open `MidiSampleScene.unity`.
2. Open `Window > MIDI > Monitor` ([Editor Tools \(MIDI Monitor\)](#)).
3. Enter Play Mode.
4. Connect a MIDI device (or use a network transport such as RTP-MIDI / UDP MIDI 2.0).
5. Confirm the following:
 - device attach events appear.
 - note and CC events are received (the monitor's IN column).
 - sending produces output on the target device (the monitor's OUT column).

Gameplay Components (MidiGameplaySampleScene)

1. Open `MidiGameplaySampleScene.unity`.
2. Enter Play Mode.
3. In the on-screen IMGUI panel, try the following:
 - **NoteOn C4 (ch0)** → fires the Router's Launch binding
 - **CC64 = 127 (ch0)** → fires the Toggle binding
 - **NoteOn C4 (ch1)** → blocked by the ChannelFilter (does not reach the Router / Tracker)
 - **Add D4 + E4 (ch0)** → 3-note chord detected by the NoteTracker
4. When a physical MIDI controller is connected, C4 / CC64 on ch0 behave the same way.
5. For details on the filter chain, see [Gameplay Components](#).

Animator Integration (MidiAnimatorIntegrationSampleScene)

1. Open `MidiAnimatorIntegrationSampleScene.unity`.
2. Enter Play Mode.
3. In the IMGUI panel, **CC1 = 127** → confirm the cube rises.
4. **Note 60** → confirm the pulse effect (scale change).
5. **Program Change 5** → confirm the `Program` Int parameter update and hue change.

6. **MIDI Start** → confirm the `TransportStart` Trigger and pulse effect.
7. **SysEx** → confirm `onMessage SysEx` appears in the Event Log (Animator parameters are not updated).
8. **CC10 / CC11** → confirm the cube rotates.
9. For details, see [Unity Ecosystem Integration](#).

Timeline Integration (MidiTimelineIntegrationSampleScene)

1. Confirm the `com.unity.timeline` package is installed.
2. Confirm `FEATURE_USE_TIMELINE` is added in Project Settings.
3. Open `MidiTimelineIntegrationSampleScene.unity`.
4. Enter Play Mode and click **Play Timeline**.
5. Confirm the Director time and SmfPlayer time stay in sync.
6. Confirm the SmfPlayer follows with **Pause / Seek to 1.0 s**.

Visual Scripting Integration (MidiVisualScriptingIntegrationSampleScene)

1. Confirm the `com.unity.visualscripting` package is installed.
2. Confirm `FEATURE_USE_VISUALSCRIPTING` is added in Project Settings.
3. Open `MidiVisualScriptingIntegrationSampleScene.unity`.
4. Enter Play Mode and confirm the cube's color changes with **NoteOn C4 / CC1 = 127**.
5. You can add a Script Machine to the same GameObject and extend the graph with **Events > MIDI** nodes.

Network MIDI (MidiNetworkJamSampleScene)

1. Add `FEATURE_MIDI_NETWORK` in Project Settings.
2. Open `MidiNetworkJamSampleScene.unity`.
3. In Play Mode, try the Hub / Client broadcast operations.

Network MIDI — Mirror / Netcode / WSNet2

1. Install the framework and enable symbols (see [Integrations — Networking](#) / Networking README).
2. Open `MidiMirrorNetworkSampleScene.unity`, `MidiNetcodeNetworkSampleScene.unity`, or `MidiWsnet2NetworkSampleScene.unity`.
3. Mirror/Netcode: **Start Host** then a second instance as **Client**. WSNet2: **Create Room** then **Join by Room#** (servers must be running).
4. Use Note / CC / Seek buttons to verify sync.

MidiClockSync / Chords & Scales (Gameplay samples)

1. Open `MidiClockSyncSampleScene.unity` and confirm Clock injection and BPM estimation in Play Mode.
2. Confirm chord/scale detection in `ChordScaleSampleScene.unity` / `ChordPuzzleSampleScene.unity` / `ScalePracticeSampleScene.unity`.
3. For details, see [Genre Kits](#) and [Gameplay Components](#).

Input System Bridge (InputSystemBridgeSampleScene)

1. Add `FEATURE_INPUT_SYSTEM` in Project Settings (`com.unity.inputsystem` is required).
2. Open `InputSystemBridgeSampleScene.unity`.
3. In Play Mode, confirm the Synthetic Device and the bidirectional bridge.

Chunity Bridge (ChunityBridgeSampleScene)

1. Install Chunity and place `Chunity.Runtime.asmdef.example` at the Scripts root as `Chunity.Runtime.asmdef`.
2. Add `FEATURE_CHUNITY` in Project Settings.
3. Open `Assets/MIDI/Samples/Integrations/Chunity/Scenes/ChunityBridgeSampleScene.unity`.
4. In Play Mode, confirm a monophonic SinOsc sounds from the Note On / Off buttons.

Chunity Workflows (ChunityWorkflowsSampleScene)

1. Follow the same enablement steps as above.
2. Open `ChunityWorkflowsSampleScene.unity`.
3. Switch between the Poly / SMF / Clock / Event / Bank / Array / Adv (HostAdvancer) / Life (PatchLifecycle) tabs and confirm each demo.

Chunity Presets (ChunityPresetsSampleScene)

1. Follow the same enablement steps as above.
2. Open `Assets/MIDI/Samples/Integrations/Chunity/Scenes/ChunityPresetsSampleScene.unity`.
3. Confirm the Soft / Bright / Pad preset switching, sliders such as Filter / Gain, and Note On.
4. (Optional) When `FEATURE_USE_TIMELINE` is enabled, demo buttons for the Timeline marker API also appear.

Chunity Mic FX (ChunityMicFxSampleScene)

1. Follow the same enablement steps as above.
2. Open `Assets/MIDI/Samples/Integrations/Chunity/Scenes/ChunityMicFxSampleScene.unity`.

3. In Play Mode, confirm mic input is enabled, and confirm that the CC1 slider or button changes the PitShift shift amount.
4. CC1 (Mod Wheel) from a physical MIDI device also reaches the same parameter via `MidiChuckBridge`.

Chunity Scriptable Generator (`ChunityGeneratorWorkflowSampleScene`)

1. In addition to the same prerequisites as the Chunity Bridge, apply the `useBuiltInAudioFilter` patch from the [Chunity Optional patch guide](#).
2. Install Unity package `com.unity.collections` if it is not already in the project.
3. Add `FEATURE_CHUNITY_SCRIPTABLE_AUDIO` in Project Settings (keep `FEATURE_CHUNITY` as well).
4. Open `ChunityGeneratorWorkflowSampleScene.unity` and confirm Main + Sub Generator + Bridge (spatialization with Note On and Near/Mid/Far).

Scriptable Audio Pipeline

1. Confirm Unity 6000.3+ and add `FEATURE_SCRIPTABLE_AUDIO` (see [Integrations](#) / [Build PostProcessing](#)).
2. Open `ScriptableAudioMetronomeSampleScene.unity`, `ScriptableAudioSequenceSampleScene.unity`, or `ScriptableAudioUmpSequenceSampleScene.unity`.
3. In Play Mode, confirm the metronome or sequence scheduling behavior.

Foundation (`FoundationSampleScene`)

1. Open `FoundationSampleScene.unity`.
2. In Play Mode, confirm device selection, latency calibration, and the Foundation UI shell.

MIDI Tracker (`MidiTrackerSampleScene`)

1. Open `Assets/MIDITracker/Scenes/MidiTrackerSampleScene.unity`.
2. Enter Play Mode and exercise pattern edit / playback / arrangement (external MIDI sound source recommended).
3. For operations and phase status, see [MIDI Tracker README](#).

If a sample doesn't receive events:

- confirm the correct platform backend is in use,
- check transport requirements (permissions on WebGL, firewall for network transports, etc.).

If you use `MidiInputRouter` or filters, also see the wiring steps in [Gameplay Components](#).

To try SMF playback/recording, you can use `SmfPlayer` / `MidiRecorder` in [SMF Tools](#).

To inspect SMF in Edit Mode, you can use the SMF preview in [Editor Tools](#).

Source code example implementations referenced by docs

The documentation pages reference these “minimal, concrete” scripts:

- `Assets/MIDI/Samples/DocumentationExamples/Midi1QuickStartExample.cs`
- `Assets/MIDI/Samples/DocumentationExamples/Midi2QuickStartExample.cs`
- `Assets/MIDI/Samples/DocumentationExamples/MpeOutputExample.cs`
- `Assets/MIDI/Samples/DocumentationExamples/SmfPlaybackExample.cs`
- `Assets/MIDI/Samples/DocumentationExamples/RtpMidiTransportExample.cs`

MPTK-related examples:

- `Assets/MIDI/Samples/DocumentationExamples/MptkVirtualOutputSinkExample.cs`
- `Assets/MIDI/Samples/DocumentationExamples/MptkToMidiManagerInputExample.cs`
- `Assets/MIDI/Samples/DocumentationExamples/MptkBootstrapExample.cs` — Bootstrap / SmfPlayer integration / validation
- `Assets/MIDI/Samples/DocumentationExamples/MptkMidiEventPipelineExample.cs` — pre-synthesis rewrite (MPTK_PRO)
- `Assets/MIDI/Samples/DocumentationExamples/MptkWriterExternalExample.cs` — Writer / External round-trip (MPTK_PRO)
- `Assets/MIDI/Samples/DocumentationExamples/MptkInnerLoopExample.cs` — InnerLoop region loop (MPTK_PRO)
- `Assets/MIDI/Samples/DocumentationExamples/MptkListPlayerExample.cs` — Playlist bridge (MPTK_PRO)
- `Assets/MIDI/Samples/DocumentationExamples/MptkSoundFontLoaderExample.cs` — Runtime SoundFont load
- `Assets/MIDI/Samples/DocumentationExamples/MptkDelayedNoteDispatcherExample.cs` — CatchMusic-style delayed dispatch
- `Assets/MIDI/Samples/DocumentationExamples/MptkFilePlayerChannelsExample.cs` — FilePlayer channel theater

MPTK sample GUI scripts (attach in Play Mode; `FEATURE_USE_MPTK` + `MPTK_PRO` where noted):

- Scene: `Assets/MIDI/Samples/MPTK/Scenes/MptkIntegrationSampleScene.unity` (requires `FEATURE_USE_MPTK`)
- Base script: `Assets/MIDI/Samples/MPTK/Scripts/MptkIntegrationSampleScene.cs`
- `Assets/MIDI/Samples/MPTK/Scripts/MptkMidiEventPipelineSample.cs` — pre-synthesis rewrite GUI
- `Assets/MIDI/Samples/MPTK/Scripts/MptkWriterExternalSample.cs` — Writer / External / Join GUI
- `Assets/MIDI/Samples/MPTK/Scripts/MptkInnerLoopSample.cs` — InnerLoop GUI
- `Assets/MIDI/Samples/MPTK/Scripts/MptkListPlayerSample.cs` — List player GUI (MPTK_PRO)

- `Assets/MIDI/Samples/MPTK/Scripts/MptkPhaseUtilitiesSample.cs` — Phase D utilities GUI
- `Assets/MIDI/Samples/MPTK/Scripts/MptkSpatializerSample.cs` — Spatializer GUI (`MPTK_PRO` + `MidiSpatializer` prefab)
- `Assets/MIDI/Samples/DocumentationExamples/MptkSpatializerExample.cs` — Spatializer host example
- `Assets/MIDI/Samples/DocumentationExamples/MptkTimelineMarkerExample.cs` — Timeline Seek / InnerLoop marker example
- Output preset:
`Assets/MIDI/Scripts/Integrations/MPTK/Presets/MptkVirtualOutput.preset.asset`

If you're using Maestro / MPTK, see:

- [Maestro / MPTK Integration](#)